SPECIAL PERMISSIONS

TYPES OF SPECIAL PERMISSIONS

1. suid permission
2. sgid permission
3. sticky bit permission
4. acl permission
5. sudo permission

SUID (super user identification permission)

SUID (Set User ID) is a special Linux permission that allows a user to run a file with the permissions of the file's owner, instead of the permissions of the user running it.

Applying SUID

1. chmod u+s (command)

Removing SUID

2. chmod u-s (command)

Example

1. ls -l /usr/bin/passwd
   -rwsr-xr-x
2. chmod u-s /usr/bin/passwd
   -rwxr-xr-x
3. su – mayur
   passwd
   error: passwd: Authentication token manipulation error
4. sudo -i
5. chmod u+s /usr/bin/passwd
6. ls -l /usr/bin/passwd
   -rwsr-xr-x 1

Then change the password, u can successfully change the password

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SGID (super group identification permission)

1. This permission can inherent the group property means the group of parent directory inherit with the child directory and files.
2. SGID (Set Group ID) is a special permission in Linux that makes files or directories inherit the group ownership of the parent — not the user who created or runs it

Applying SGID
chmod g+s "directory/file name”

Removing SGID
chmod g-s “directory/file name”

1.create a group

1. groupadd tech
2. passwd tech
3. gpasswd -M user1,user2 tech

2.create a directory

1. mkdir /project
2. ls -ld /project
   rwx r-x r-x
3. chmod 770 /project
   ls -ld /project
4. rwx rwx ---

3.add group into the project directory

1. chgrp tech /project
2. ls -ld /project
   rwx rwx --- root tech project

4. assin sgid permission to a /project file

1. chmod g+s /project
2. ls -ld /project
   rwx rws --------root tech ------s denotes there is a execution permission

5. To create file in /project

1. su - user1
2. cd /project
3. touch file1

9.To create file in /project

1. su - user2
2. cd /project
3. touch file2

STICKY BIT PERMISSION

1. This permission basically used for controlling all the users to resist the delete and modify operation.
2. only the owner and the root user have to access to delete or modifies the file.

Applying Sticky bit permission
chmod o+t "file name"

Removing Sticky bit permission
chmod o-t "file name

1.applying the sticky bit permission

1. chmod o+t /project
2. ls -ld /project
   rwx rws --T

note:

1. T----------> there is not execution permission
2. t----------> there is a execution permission

ACL (access control list) PERMISSION

it is a special type of permission that adds extra permission to a user to use particular directory or file and also denies all other users to use that directory.

In short

1. particular user high access
2. particular user deny access

HOW TO APPLY ACL

1. mkdir /test
2. ls -ld /test
   rwx r-x r-x
3. setfacl -m u:mayur:rwx /test

4. getfacl /test
   user:mayur:rwx
5. su – mayur
6. cd /test
7. touch file

Note:
Before applying acl mayur user cannot create the file because there is no write permission mayur user can only list and execute the file

HOW TO DENY OTHERS USER PERMISSION

1. setfacl -m u:user1:--- /test
2. getfacl /test
   user:user1:---
3. su – user1
4. cd /test
   -bash: cd: /test: Permission denied

HOW TO APPLY ACL ON GROUP

1. setfacl -m g:tech:rwx /test
2. getfacl /test
   group:tech:rwx

HOW TO REMOVE ACL

1. setfacl -x u:mayur /test

HOW TO REMOVE ALL ACL

1. setfacl -b /test
